

Assessing Bicycle Lane Quality Using Smartphone Sensor Data

Group 11

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1. Business / Objective Understanding

Poorly maintained bicycle lanes, with cracks, potholes and uneven paving, reduce cyclist comfort and safety and can discourage cycling, an everyday form of physical activity with clear health benefits. Manual inspection of cycling infrastructure is slow and difficult to scale.

This project investigates whether the accelerometer, gyroscope and gravity sensors of a smartphone mounted on a bicycle can distinguish smooth from bumpy bicycle lanes. The objective is to compare supervised and unsupervised approaches, select one model, and apply it to an independent external ride to map its smooth and bumpy sections: the basis for a low-cost tool that could help city planners prioritize maintenance.

2. Data Understanding

2.1 Dataset and Data Collection

Three participants recorded data with the Sensor Logger app (accelerometer, gyroscope and gravity at 100 Hz). The phone was fixed to the handlebar to keep its orientation consistent, participants cycled at about 15–18 km/h, and each recording lasted about 60 seconds. Each participant, with their own phone (iPhone 12, iPhone 15 or iPhone 15 Pro Max) and bicycle, recorded five rides on a smooth lane (a well-maintained red cycling lane) and five on a bumpy lane (noticeable but safe irregularities such as cracks and uneven paving). This gives **30 recordings, 15 per surface**. Because each phone was used by one rider on one bicycle, the effects of phone, rider and bicycle cannot be separated.

2.2 Exploratory Data Analysis

For each sensor, the magnitude $|v| = \sqrt{x^2 + y^2 + z^2}$ removes the dependence on the phone's orientation. On every phone, bumpy rides show clearly higher acceleration and rotation magnitudes than smooth rides (Figure 1a). The second finding is a strong **device effect**: the overall level differs several-fold between phones, so that a smooth ride on the iPhone 15 Pro Max vibrates more than a bumpy ride on the iPhone 12. A model trained on raw values would partly learn which phone recorded a ride.

2.3 Data Quality and Limitations

- Small irregularities in the sampling timestamps, and non-cycling movement at the start and end of each recording.
- Strong differences in sensor response between the phones (Section 2.2).
- A small dataset: 30 rides from three riders. The 1,389 windows extracted from them overlap and are far from independent.
- Labels per ride: a “smooth” ride may contain an isolated bump, and a “bumpy” ride calmer stretches.

3. Data Preparation

3.1 Preprocessing

The first and last 10 seconds of each recording were removed to exclude the handling of the phone, and all signals were resampled onto a regular 100 Hz grid.

The train/test split was made next, **before any step that is fitted to the data**, so that neither the normalization nor the feature selection could be influenced by the held-out rides. The split is over recordings, not windows: 24 rides for

training and 6 for testing, stratified so that the test set contains exactly one ride of every combination of surface and phone.

Each magnitude was then standardized per phone, with the mean and standard deviation computed from that phone's training rides only. This removes the device effect while preserving the difference between surfaces: after normalization, the two surfaces no longer overlap across phones (Figure 1b).

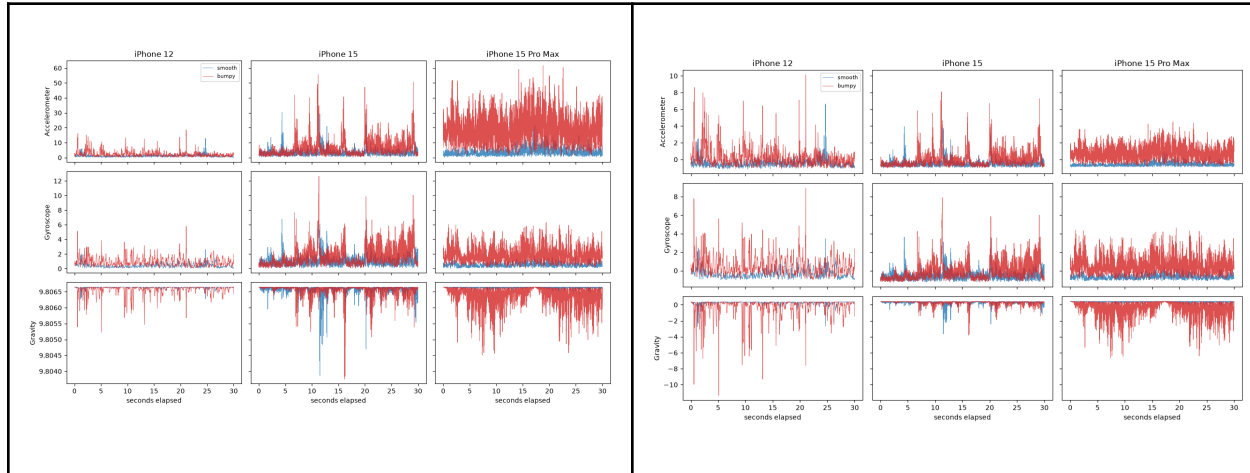


Figure 1. Magnitudes of one smooth (blue) and one bumpy (red) ride per phone, (a) before and (b) after per-phone normalization.

3.2 Feature Engineering and Extraction

Recordings were divided into 2-second windows with 50% overlap (about 200 samples each), giving 1,389 windows. Each window is described by:

- five statistics (mean, median, standard deviation, minimum and maximum) of the acceleration, rotation and gravity magnitudes: 15 features;
- the energy of the acceleration and rotation magnitudes in six frequency bands (0.5–2, 2–4, 4–8, 8–16, 16–32 and 32–50 Hz), computed with a discrete Fourier transform, to test whether road vibration and the rider's own motion occupy different frequencies: 12 features.

This gives **27 candidate features**.

3.3 Feature Selection

Each feature was ranked by mutual information, information gain and point-biserial correlation with the label (Figure 2). Features revealing more than 10% of the phone's identity were excluded as device fingerprints, features correlating less than 0.40 with the label were dropped, and features correlating above 0.90 with a better-ranked one were removed as redundant. The main findings:

- The mean and median of the acceleration and rotation magnitudes are the most informative features (about 0.58 bits each); the best one alone removes about 59% of the uncertainty about the label.
- Road vibration appears below about 8 Hz. These bands carry 0.35–0.48 bits and reveal little about the phone. Above 16 Hz the bands mostly identify the phone (15–61% of its identity), because high-frequency vibration depends on the phone, its mount and the bicycle more than on the road.
- Several features are pure device fingerprints; the maximum of the gravity magnitude identifies the phone completely.

Seven features were selected: the mean acceleration and rotation magnitudes (acc_mag_mean, gyro_mag_mean) and the band energies of acceleration at 0.5–2, 2–4 and 4–8 Hz and of rotation at 0.5–2 and 2–4 Hz.

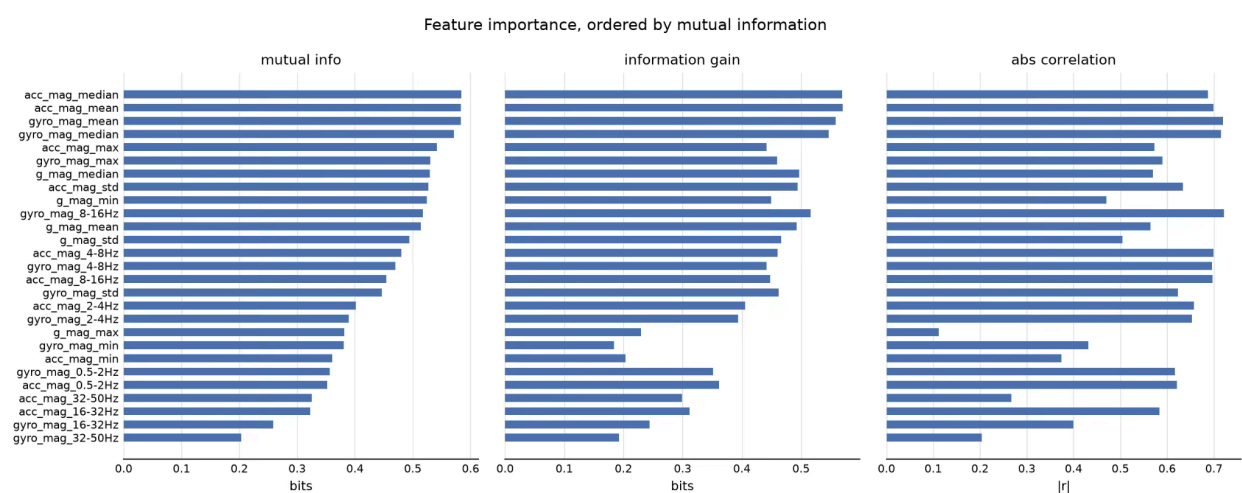


Figure 2. Importance of the 27 candidate features under the three measures, ordered by mutual information.

4. Modeling

Four unsupervised and four supervised models were trained on the seven selected features.

4.1 Unsupervised Models

One method from each main clustering family was used: **K-means** (partitioning), **DBSCAN** (density-based), a **Gaussian mixture** (probabilistic) and **agglomerative clustering with Ward linkage** (hierarchical). Each was asked for two groups without seeing the labels, on individual windows and on the features averaged per recording; DBSCAN's eps was chosen, also without labels, as the value giving two clusters with the least noise. The labels were used only for evaluation (Table 1, Figure 3).

Table 1. Unsupervised models. ARI: Adjusted Rand Index against the surface; agreement: share of samples whose cluster's majority label is their own.

Model	Windows: ARI	Windows: agreement	Windows: silhouette	Recordings: ARI
K-means	0.542	86.8%	0.495	0.867
DBSCAN	0.313	27.6% (70% noise)	0.708	0.806
Gaussian mixture	0.621	89.4%	0.435	0.867
Agglomerative (Ward)	0.590	88.4%	0.488	0.867

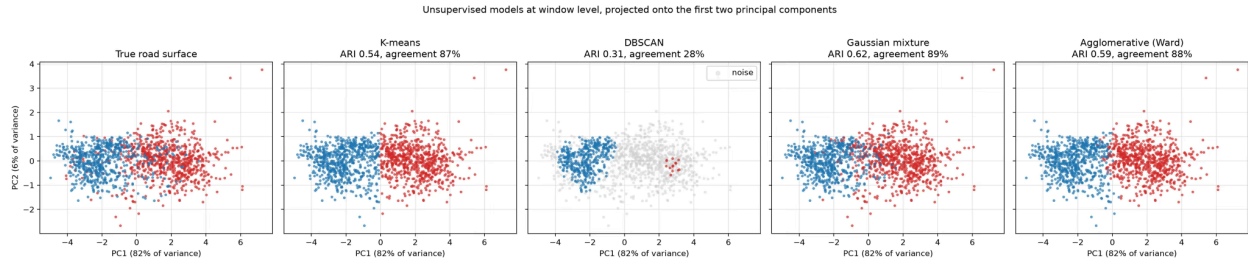


Figure 3. Window-level clusters projected onto the first two principal components, each cluster coloured by its majority surface.

4.2 Supervised Models

Logistic regression, a **random forest**, **XGBoost** and an **SVM with an RBF kernel** were trained as pipelines in their default configuration (with standardization inside the pipeline where the model needs it). They were evaluated with 5-fold cross-validation over the 24 training rides, grouped by recording so that no ride appears on both sides of a fold (Table 2).

Table 2. Supervised models: accuracy on the training windows and in 5-fold grouped cross-validation.

Model	Training accuracy	Cross-validated accuracy
Random forest	1.000	0.884 ± 0.054
SVM (RBF)	0.906	0.879 ± 0.054
Logistic regression	0.899	0.875 ± 0.064
XGBoost	1.000	0.842 ± 0.059

4.3 Model Selection and Hyperparameter Tuning

The final model was chosen with the **one-standard-error rule**. The best cross-validated accuracy (random forest, 0.884) has a standard error of 0.024, so the random forest, the SVM and logistic regression are statistically indistinguishable, and the simplest of them, **logistic regression**, was selected (justified in Section 5.2).

Its regularization strength C was tuned by grid search (13 values from 0.001 to 1000) over the same grouped folds. The grid is flat, with every value within one standard error of the best, and repeating the search with ten different fold assignments produced four different winners. Since a tuned value replaces the default only if it beats it by more than one standard error, **the default $C = 1$ was kept**.

5. Evaluation

5.1 Validation Strategy

Windows overlap, so a random split over windows would put near-duplicates in both training and test data. All validation is therefore by recording: the held-out split described in Section 3.1, and cross-validation grouped by recording for model selection and tuning. The six test rides were used only once, for the final evaluation. Classifiers are assessed with accuracy, balanced accuracy, precision, recall, F1-score, ROC AUC and the confusion matrix; clusterings with the ARI and label agreement against the surface, the silhouette score, and the ARI against the phone, which checks that clusters do not simply separate the phones.

5.2 Model Comparison

Unsupervised. The Gaussian mixture recovers the surface best at window level (ARI 0.621, 89.4% agreement), as it models the overlapping, elongated groups that the correlated features form; Ward and K-means follow. DBSCAN reaches two clusters only by declaring 70% of the windows noise, isolating the dense core of smooth riding, yet it has the highest silhouette score: internal cluster quality is no guide to meaning here. At recording level, three methods place 29 of the 30 rides correctly. The ARI against the phone is near zero throughout, confirming that the normalization removed the device effect.

Supervised. The random forest, the SVM and logistic regression differ by less than 0.01, far less than the variation between folds; only XGBoost falls clearly behind. Both tree ensembles fit the training windows perfectly without generalizing better.

Supervised against unsupervised. The best clustering agrees with the labels about as often (89.4%) as the best classifier is correct (0.884), but it is scored on the windows it was fitted to and needs labels to name its clusters, whereas the classifier is scored on unseen rides. Supervised learning gives a validated probability per window but needs labelled rides; unsupervised learning needs no labels and reveals structure, but cannot name its clusters and depends on parameters and metrics that can mislead. Clustering suits exploring unlabelled routes or phones and flagging unusual stretches; classification suits routine use once labelled rides exist.

Logistic regression is the optimal model by the five criteria of the assignment. **Problem suitability:** the surfaces differ mainly in vibration and rotation strength, a nearly linear boundary, and the model gives a probability per 2-second stretch. **Data requirements:** a low-capacity model suits 24 training rides, which the tree ensembles memorize. **Interpretability:** one weight per feature makes it a transparent roughness score. **Efficiency:** seven multiplications per window, trivial on a phone. **Performance:** within one standard error of the best cross-validated model.

5.3 Final Results and Error Analysis

On the six held-out rides, logistic regression reaches an **accuracy of 0.932**, a **balanced accuracy of 0.931** and a **ROC AUC of 0.985** (precision and recall: smooth 0.914 and 0.929, bumpy 0.945 and 0.934). Of the 293 test windows, 9 smooth windows were classified as bumpy and 11 bumpy windows as smooth (Figure 4). The test set is easier than the average fold, so the cross-validated 0.875 is the more representative estimate.

Errors arise where the surfaces vibrate alike: milder stretches of a bumpy lane, and isolated bumps on a smooth lane that, given the ride-level labels, may correctly detect rough spots; speed and riding style add to this. Some folds are hard for every model (0.74–0.82, against 0.88–0.95), so part of the error belongs to particular recordings rather than to the classifier.

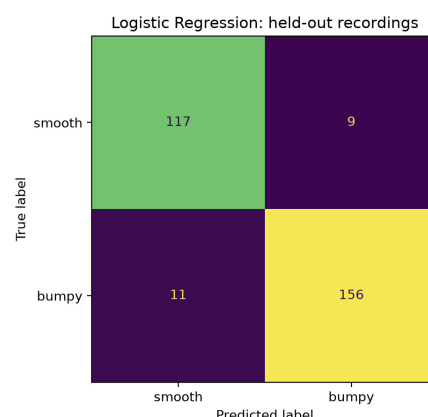


Figure 4. Confusion matrix of the final model on the held-out rides.

6. Deployment Considerations and Recommendations

6.1 External Dataset

The final model was applied to the independent external ride (16.7 minutes after trimming), processed with the same pipeline. Its phone is unknown, so the per-phone normalization cannot be applied. Two normalizations using only the ride itself were compared on simulated rides, built by joining held-out smooth and bumpy recordings in known proportions (Figure 5): standardizing by the ride's average, which assumes a balanced mix, and **comparing each window with the ride's quietest 5% of windows**, which only assumes some smooth road. The latter is clearly better on mostly smooth rides (75% against 31% correct on an entirely smooth ride) and was used; it fails only on a ride bumpy throughout, and flags about 25% of an entirely smooth ride as bumpy. The percentile was chosen on training rides, but the choice of normalization was informed by the external ride, in which the ride average split roughly in half.

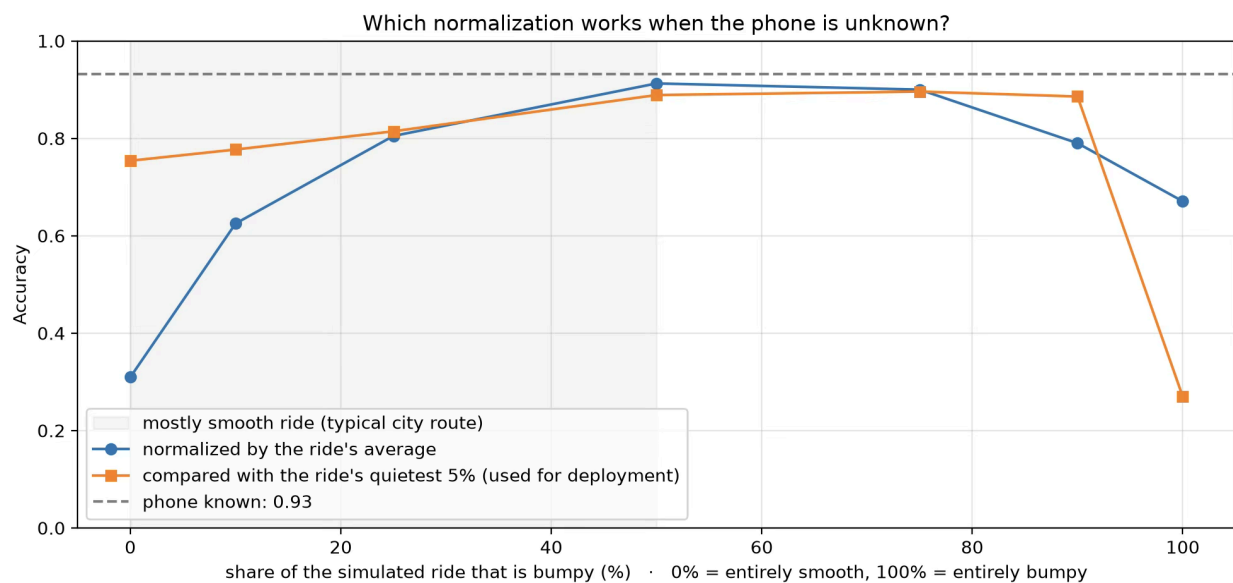


Figure 5. Accuracy on simulated held-out rides with the phone treated as unknown, against the share of the ride that is bumpy.

The model classifies **11.4% of the ride's windows as bumpy** (12.3% while moving), in 59 short sections of at most 5 seconds, several of them at the largest spikes in the signal. The two stops are classified smooth (Figure 6). This is well below the false-alarm rate of an entirely smooth ride, so the ride is best described as **a smooth lane with a few brief rough spots**. The share depends on the reference: the 10th percentile, practically tied with the 5th on training rides, gives 2.0%, so the bumpy share is uncertain within roughly 2–12%, but the ride is mostly smooth under both.

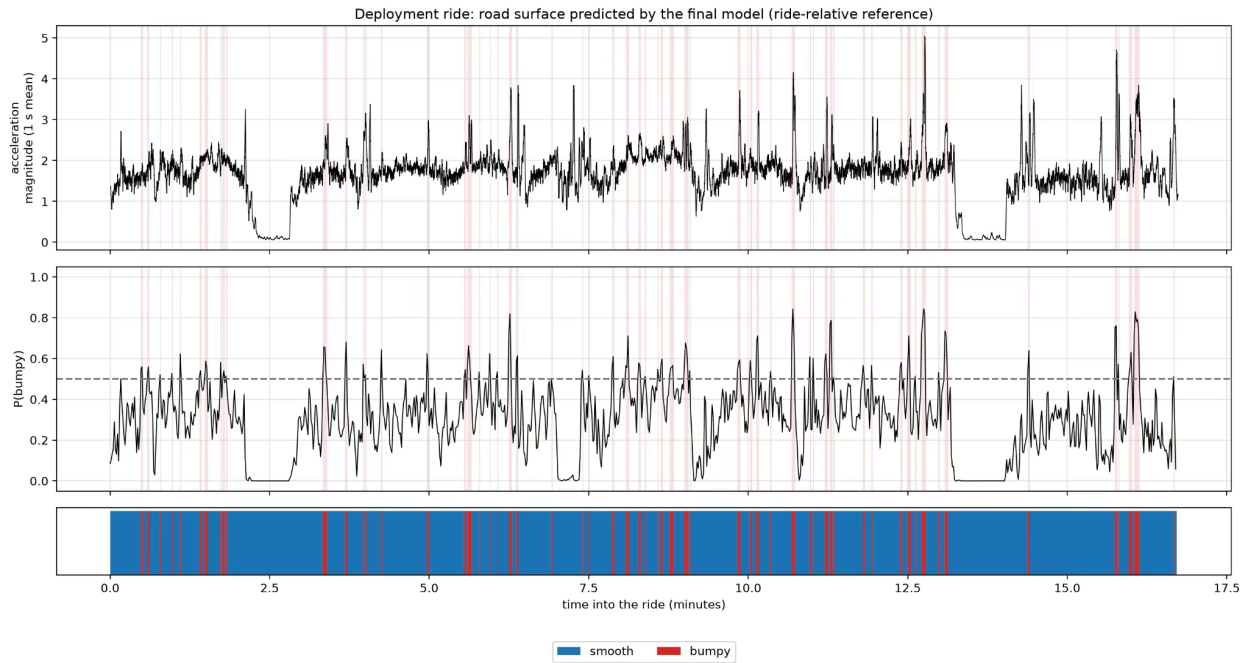


Figure 6. External ride: acceleration magnitude, predicted probability of bumpy road, and the resulting smooth/bumpy map.

6.2 Practical Considerations and Limitations

- There is no ground truth for the external route, and without a reference for the phone only relative roughness within a ride can be identified, assuming it contains some smooth road.
- Braking and pulling away around stops, unseen in our steady training rides, and a different phone mount change the vibration the sensor records.
- Phone, rider and bicycle are confounded (three of each), and labels are per ride.

6.3 Future Improvements

A short calibration ride on a known smooth lane with the deployment phone would give the model an absolute reference, and GPS would map rough spots to road positions and recognize slow riding. A ground-truth route, more riders, phones and bicycles, evaluation on an unseen phone, and per-segment labels would make the results more general.

7. Conclusion

Smartphone motion sensors distinguish smooth from bumpy bicycle lanes well once device differences are removed; the mean vibration and rotation strength and their low-frequency band energies are the most informative features. Of four supervised and four unsupervised models, logistic regression, the simplest model statistically equivalent to the best, reached an accuracy of 0.932 on unseen rides (0.875 in cross-validation). For an unknown phone a ride-relative normalization is needed, under which the external ride is mostly smooth with a few brief rough spots; a short calibration ride would make this reading absolute.